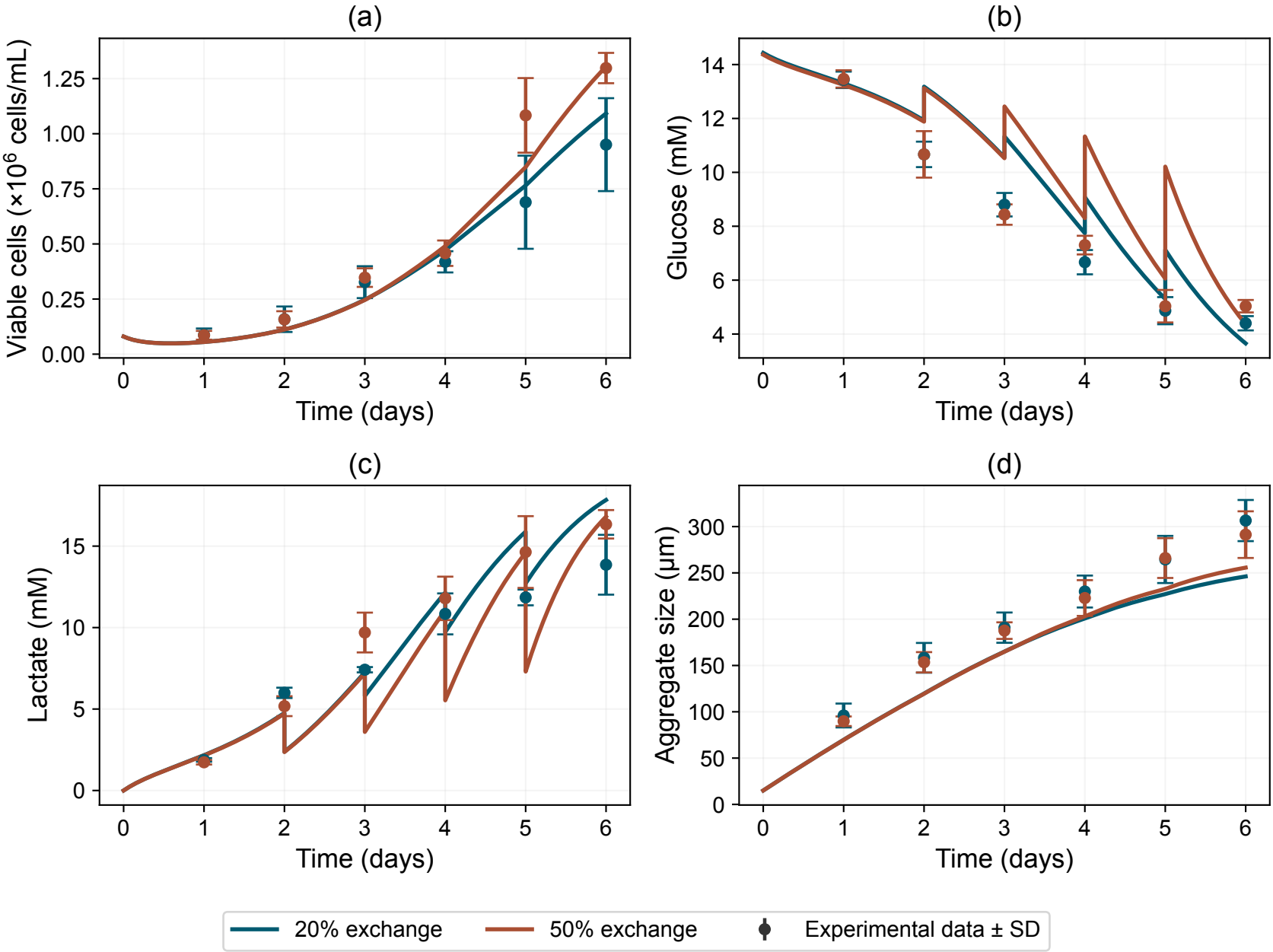
(a)
Viable cells (×10^6 cells/mL)
0.00
0.25
0.50
0.75
1.00
1.25
0
1
2
3
4
5
6
Time (days)
(b)
Glucose (mM)
4
6
8
10
12
14
0
1
2
3
4
5
6
Time (days)
(c)
Lactate (mM)
0
5
10
15
0
1
2
3
4
5
6
Time (days)
(d)
Aggregate size (µm)
0
50
100
150
200
250
300
0
1
2
3
4
5
6
Time (days)
20% exchange
50% exchange
Experimental data ± SD